COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212	**SECTION:**	CPE31S2
DATE PERFORMED:		**DATE SUBMITTED:**	
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Hands-On Activity 6.1: Targeting Specific Nodes

1. PROCEDURE OUTPUT

What procedures did you perform?

Task 1: Targeting Specific Nodes

1. Create a new playbook and named it site.yml. Follow the commands as shown in the image below. Make sure to save the file and exit.
2. Edit the inventory file. Remove the variables we put in our last activity and group according to the image shown below:

Make sure to save the file and exit.

Right now, we have created groups in our inventory file and put each server in its own group. In other cases, you can have a server be a member of multiple groups, for example you have a test server that is also a web server.

3. Edit the site.yml by following the image below:

Make sure to save the file and exit.

The pre-tasks command tells the ansible to run it before any other thing. In the pre-tasks, CentOS will install updates while Ubuntu will upgrade its distribution package. This will run before running the second play, which is targeted at web_servers. In the second play, apache and php will be installed on both Ubuntu servers and CentOS servers.

Run the site.yml file and describe the result.

4. Let's try to edit again the site.yml file. This time, we are going to add plays targeting the other servers. This time we target the db_servers by adding it on the current site.yml. Below is an example: (Note add this at the end of the playbooks from task 1.3.

Make sure to save the file and exit.

Run the site.yml file and describe the result.

5. Go to the remote server (Ubuntu) terminal that belongs to the db_servers group and check the status for mariadb installation using the command: systemctl status mariadb. Do this on the CentOS server also.

Describe the output.

6. Edit the site.yml again. This time we will append the code to configure installation on the file_servers group. We can add the following on our file.

Make sure to save the file and exit.

Run the site.yml file and describe the result.

The testing of the file_servers is beyond the scope of this activity, and as well as our topics and objectives. However, in this activity we were able to show that we can target hosts or servers using grouping in ansible playbooks.

```
ace@workstation:~/ansible-lab$ ansible-inventory -i inventory --graph
@all:
  |--@ungrouped:
  |--@web_servers:
  |  |--192.168.56.105
  |  |--192.168.56.106
  |--@db_servers:
  |  |--192.168.56.108
  |--@file_servers:
  |  |--192.168.56.107
ace@workstation:~/ansible-lab$
```

Task 2: Using Tags in running playbooks
In this task, our goal is to add metadata to our plays so that we can only run the plays that we want to run, and not all the plays in our playbook.
1. Edit the site.yml file. Add tags to the playbook. After the name, we can place the tags: name_of_tag. This is an arbitrary command, which means you can use any name for a tag.
Make sure to save the file and exit.
Run the site.yml file and describe the result.
2. On the local machine, try to issue the following commands and describe each result:
2.1 ansible-playbook --list-tags site.yml
2.2 ansible-playbook --tags centos --ask-become-pass site.yml
2.3 ansible-playbook --tags db --ask-become-pass site.yml
2.4 ansible-playbook --tags apache --ask-become-pass site.yml
2.5 ansible-playbook --tags “apache,db” --ask-become-pass site.yml

```
ace@workstation:~$ ansible-playbook --list-tags site.yml
[WARNING]: No inventory was parsed, only implicit localhost
 is available
[WARNING]: provided hosts list is empty, only localhost is
available. Note that
the implicit localhost does not match 'all'
[WARNING]: Could not match supplied host pattern, ignoring:
 web_servers
[WARNING]: Could not match supplied host pattern, ignoring:
 db_servers

playbook: site.yml

  play #1 (all): Update all servers     TAGS: []
      TASK TAGS: [centos]

  play #2 (web_servers): Configure web servers  TAGS: []
      TASK TAGS: [apache]

  play #3 (db_servers): Configure database servers      TAG
S: []
      TASK TAGS: [db]
ace@workstation:~$
```

Task 3: Managing Services

1. Edit the file site.yml and add a play that will automatically start the httpd on CentOS server.

Figure 3.1.1

Make sure to save the file and exit.

You would also notice from our previous activity that we already created a module that runs a service.

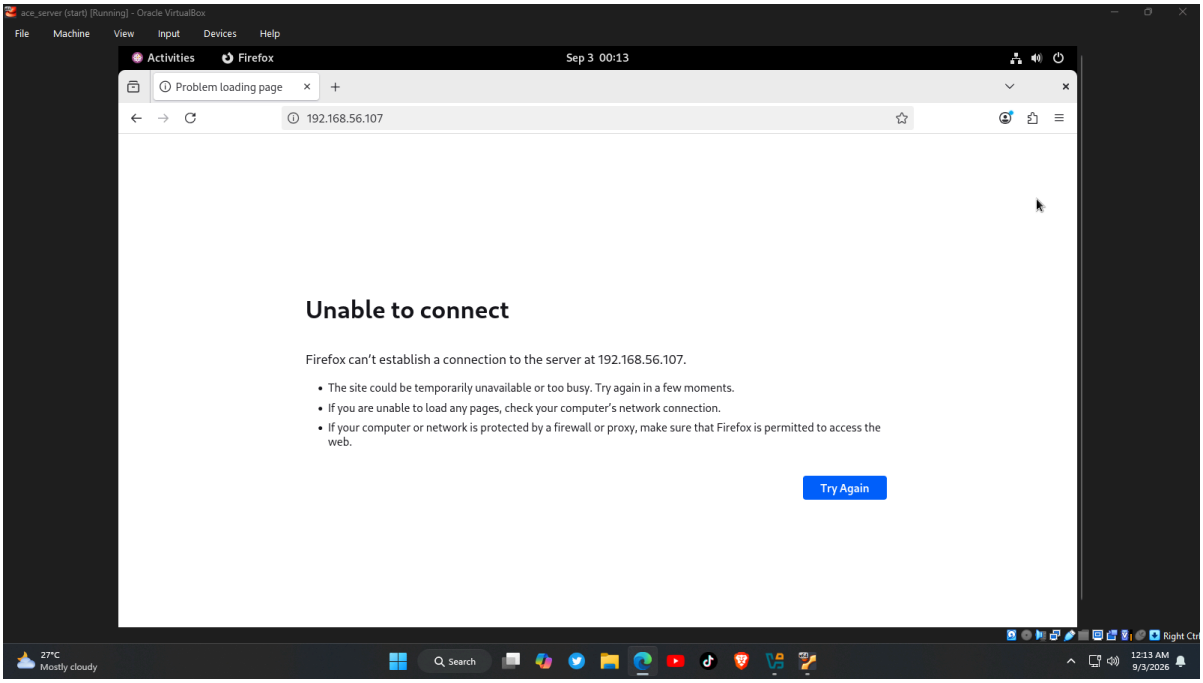
Figure 3.1.2

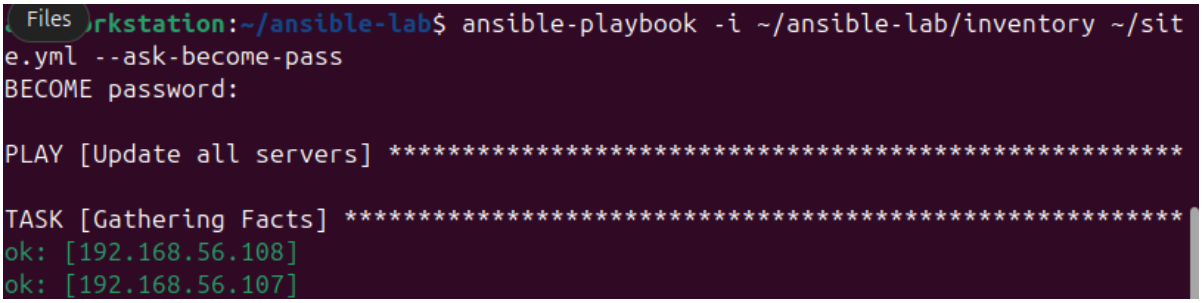
This is because in CentOS, installed packages' services are not run automatically. Thus, we need to create the module to run it automatically.

2. To test it, before you run the saved playbook, go to the CentOS server and stop the currently running httpd using the command sudo systemctl stop httpd. When prompted, enter the sudo password. After that, open the browser and enter the CentOS server's IP address. You should not be getting a display because we stopped the httpd service already.

3. Go to the local machine and this time, run the site.yml file. Then after running the file, go again to the CentOS server and enter its IP address on the browser.

Describe the result.
To automatically enable the service every time we run the playbook, use the command enabled: true similar to Figure 7.1.2 and save the playbook.





2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Reflections:
Answer the following:
1. What is the importance of putting our remote servers into groups?
   - Putting remote servers into groups makes it easier to manage a multiple tasks such as webservers or database servers, instead of all servers at once. This helps keep configurations organized, reduces the risk of applying the wrong changes to the wrong environment, and makes automation more efficient and scalable as the number of servers grows.

2. What is the importance of tags in playbooks?
   - Tags are important in playbooks since they are able to allow specific tasks or sections to be run independently instead of executing the entire playbook every time. Makes testing and debugging faster, since only the relevant parts need to be rerun.
3. Why do think some services need to be managed automatically in playbooks?
   - Some services need to be managed automatically because manually restarting or reloading them after every configuration change is very inefficient and time consuming. Automating this process ensures that services always updated,reliable, reducing human error and keeping systems running smoothly.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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